

PicoCTF 2021 – Nice netcat...

Nice netcat...

General Skills Easy 15 pts

by syreal

There is a nice program that you can talk to by using this command in a shell:

The challenge asks us to communicate with a service using the Netcat command

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ASCII Decoding

```
└─$ nc wily-courier.picoctf.net 56511  
112  
105  
99  
111  
67
```

When we use netcat to connect to the server, we receive a series of decimal numbers

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ASCII Decoding

```
└─$ nc wily-courier.picoctf.net 56511
112
105
99
111
67
```

Next, we need to decode these numbers from their current form to ASCII (text) characters

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ASCII Decoding

ASCII Encoding

ASCII	decimal	hexadecimal
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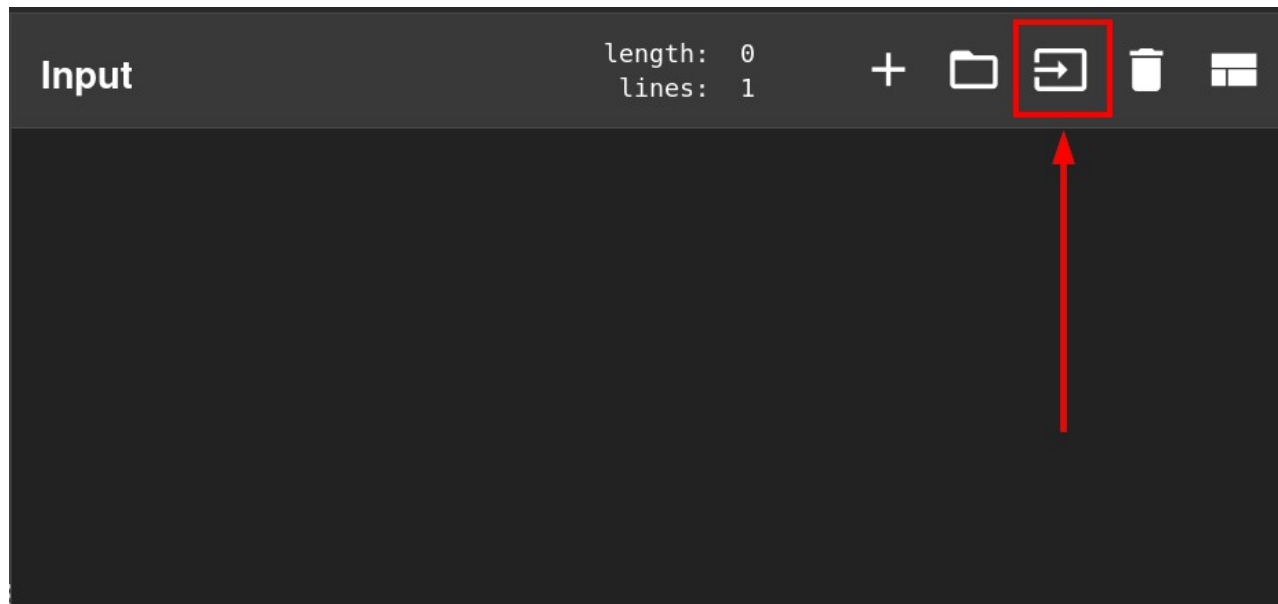
A	65	41
---	----	----

a	97	61
---	----	----

ASCII is a common text encoding system used by computing devices, and each character can be represented by a number

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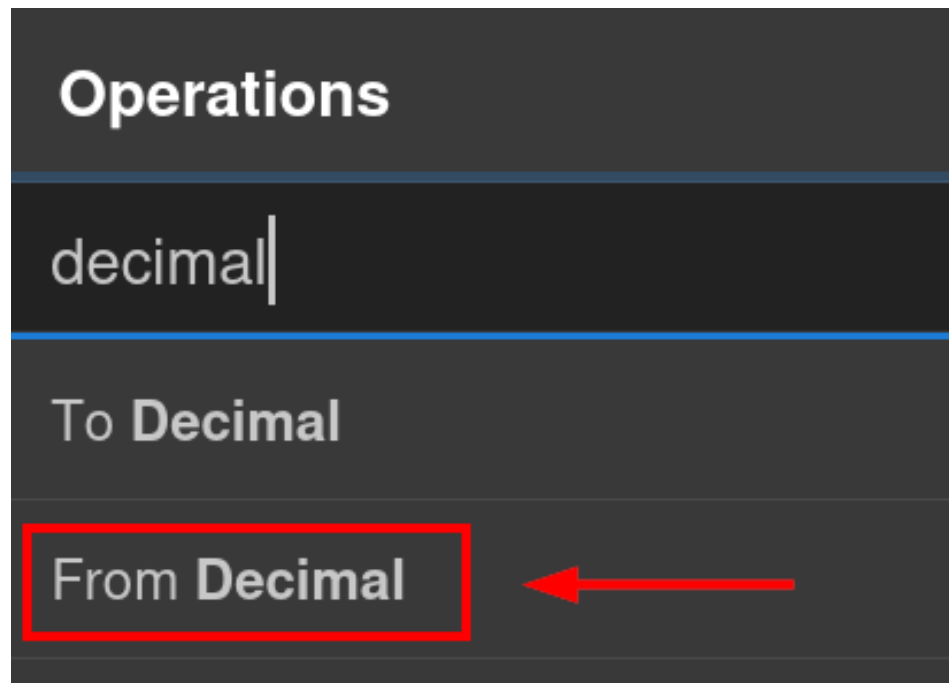
CyberChef Solve Method



To solve this challenge, using CyberChef, we first need to save the numbers as a file, then upload it as input in the `Input` window using the `Open file as input` button

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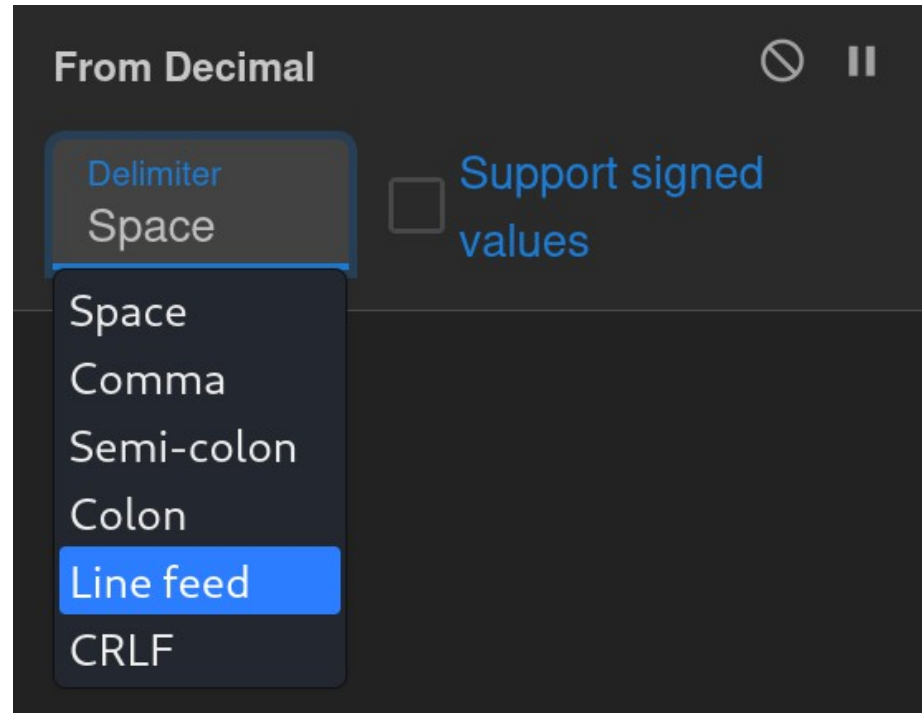
CyberChef Solve Method



Then search for the `From Decimal` operation in the `Operations` window, then drag that into the `Recipe` window

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CyberChef Solve Method



Because the numbers are separated by lines, we need to click on the `Delimiter` button and select `Line feed`

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Linux Solve Method

```
awk          ← the command itself
'{ printf "%c", $1 }' ← print the ASCII on the line
output.txt   ← the file to be affected
```

```
└─$ awk '{ printf "%c", $1 }' output.txt
picoCTF{p02d_klttyl_n1c3_klttyl_d9d00}
```

To solve this using Linux, we'll use the `awk` command

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Python Solve Method

```
with open("output.txt", "r") as f:
    num_list = f.read().splitlines("\n")

chars = ""

for i in num_list:
    chars = chars + (chr(int(i)))

print(chars)
```

To solve this using Python, we'll use `chr` function to convert numbers to their character equivalent